VHDL Simulation Assignment 2

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Problem 1. 4-input AND / OR

out1 = a and b and c and d, out2 = a or b or c or d.

Source: and_or4.vhd

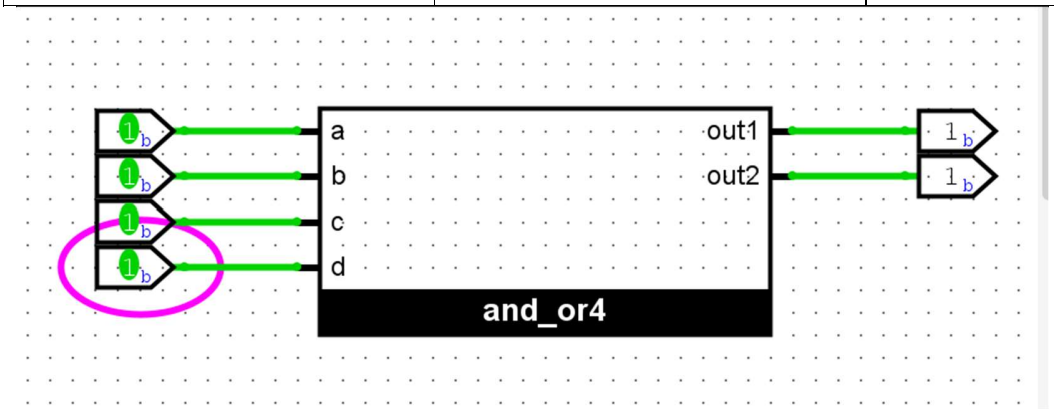
```
library ieee;
use ieee.std_logic_1164.all;

entity and_or4 is
    port (
        a, b, c, d : in std_logic;
        out1, out2 : out std_logic
    );
end entity and_or4;

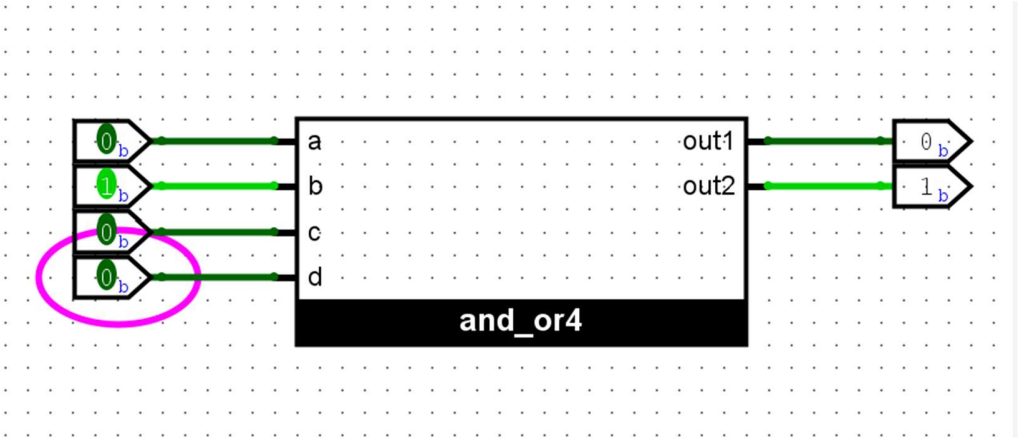
architecture bhv of and_or4 is
begin
    out1 <= a and b and c and d;
    out2 <= a or b or c or d;
end architecture bhv;
```

Problem 1 - Test results

a b c d	out1	out2
1 1 1 1	1	1
0 1 0 0	0	1



a=b=c=d=1 -> out1=1, out2=1



a=0, b=1, c=0, d=0 -> out1=0, out2=1

Problem 2. 8-to-1 Multiplexer

ch (3 bits) selects one bit of d (8 bits) as output.

Source: mux8_1.vhd

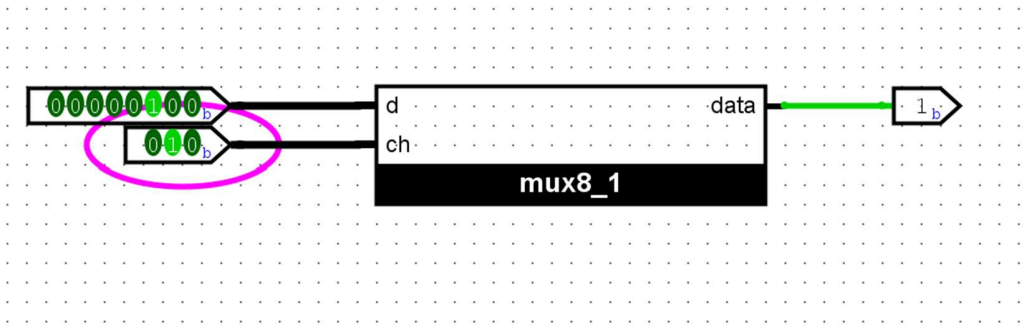
```
library ieee;
use ieee.std_logic_1164.all;

entity mux8_1 is
    port (
        d    : in  std_logic_vector(7 downto 0);
        ch   : in  std_logic_vector(2 downto 0);
        data : out std_logic
    );
end entity mux8_1;

architecture bhv of mux8_1 is
begin
    process(d, ch)
    begin
        case ch is
            when "000" => data <= d(0);
            when "001" => data <= d(1);
            when "010" => data <= d(2);
            when "011" => data <= d(3);
            when "100" => data <= d(4);
            when "101" => data <= d(5);
            when "110" => data <= d(6);
            when "111" => data <= d(7);
            when others => data <= '0';
        end case;
    end process;
end architecture bhv;
```

Problem 2 - Test results

d	ch	data
00000100	010	1



d=00000100, ch=010 -> data=1

Problem 3. 7-segment letters

Display A, b, C, d, F, g on a 7-segment display. display(7)=a ... display(0)=dp.

Source: letters.vhd

```
library ieee;
use ieee.std_logic_1164.all;

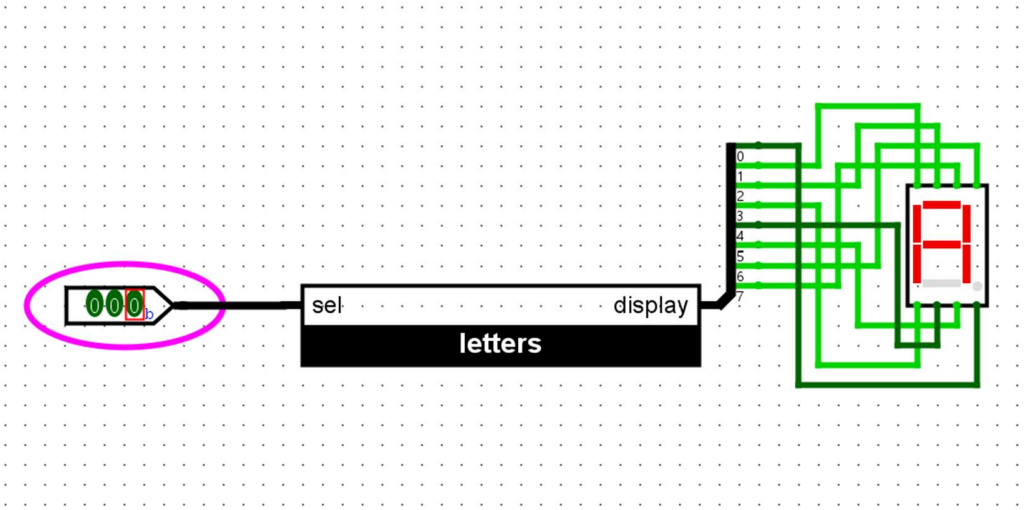
entity letters is
    port (
        sel     : in  std_logic_vector(2 downto 0);
        display : out std_logic_vector(7 downto 0)
    );
end entity letters;

architecture bhv of letters is
begin
    process(sel)
    begin
        case sel is
            when "000" => display <= "11101110"; -- A
            when "001" => display <= "00111110"; -- b
            when "010" => display <= "10011100"; -- C
            when "011" => display <= "01111010"; -- d
            when "100" => display <= "10001110"; -- F
            when "101" => display <= "11110110"; -- g
            when others => display <= "00000000";
        end case;
    end process;
end architecture bhv;
```

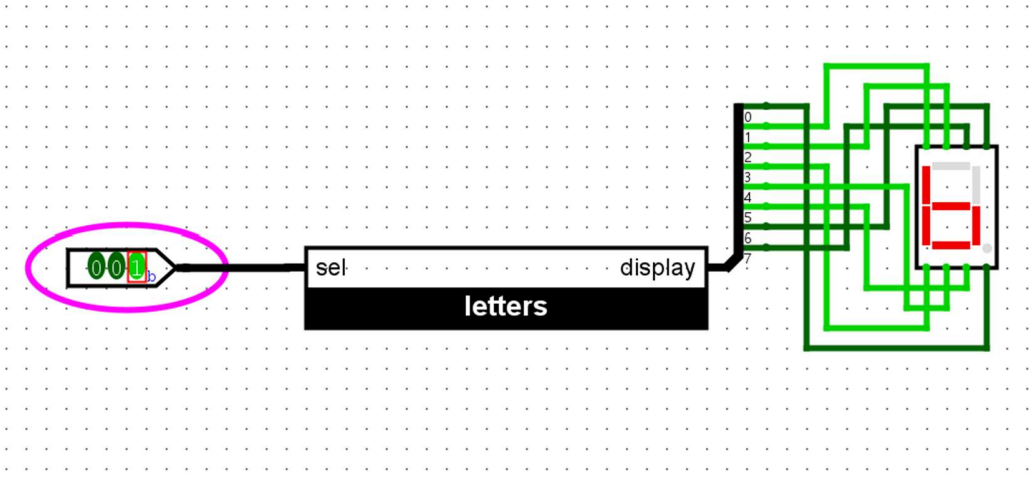
Problem 3 - Test results

sel	Letter	display (a..dp)
000	A	11101110
001	b	00111110
010	C	10011100
011	d	01111010
100	F	10001110
101	g	11110110

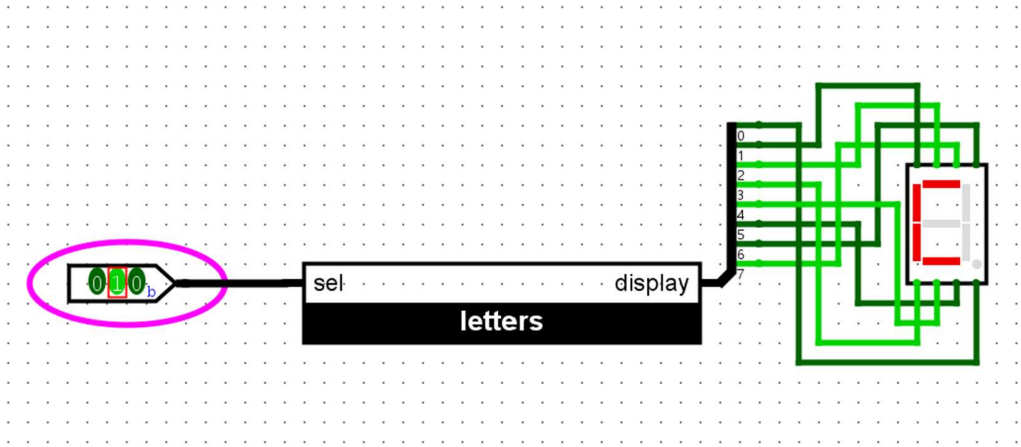
The task lists a, b, c, d, f, g; these were interpreted as the letters A, b, C, d, F, g.



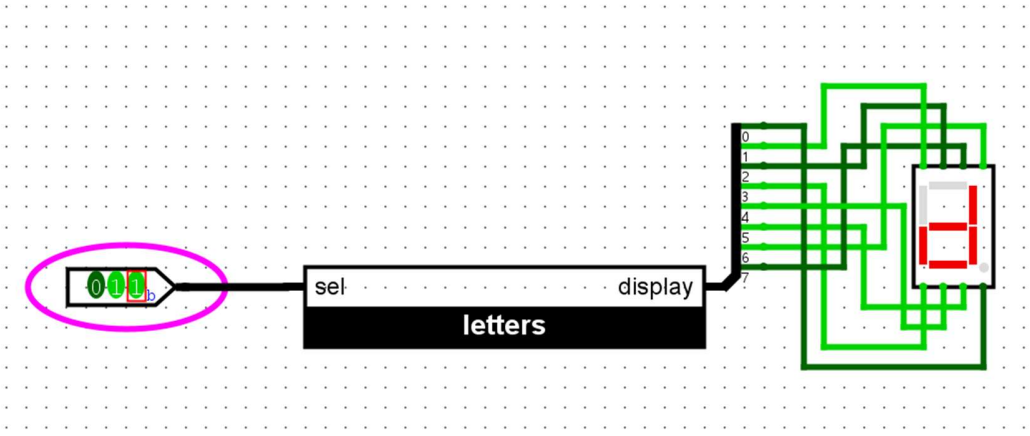
sel=000 -> A



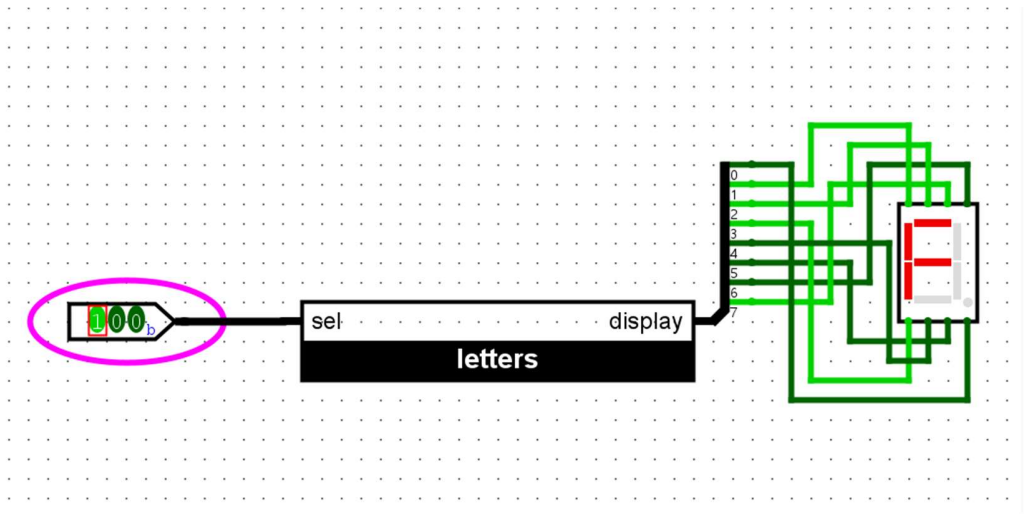
sel=001 -> b



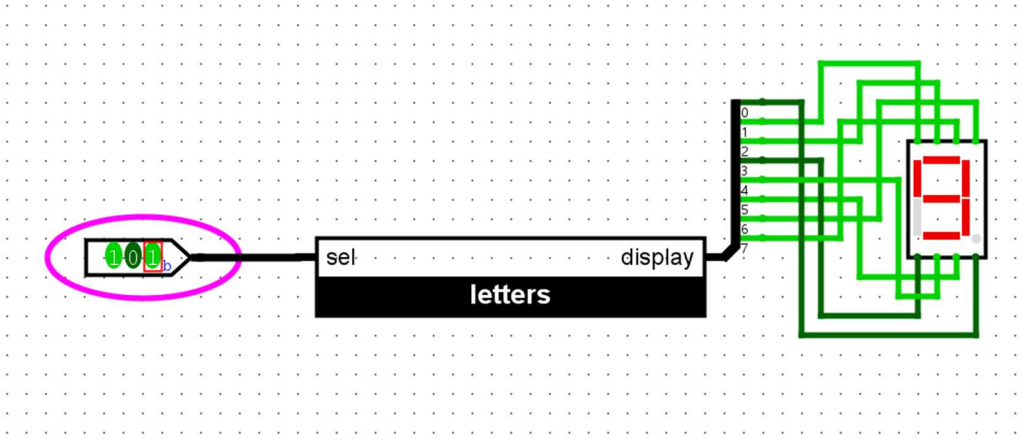
sel=010 -> C



sel=011 -> d



sel=100 -> F



sel=101 -> g

Problem 4. 25-person voting machine

result = 1 when 13 or more of the 25 inputs are 1.

Source: vote25.vhd

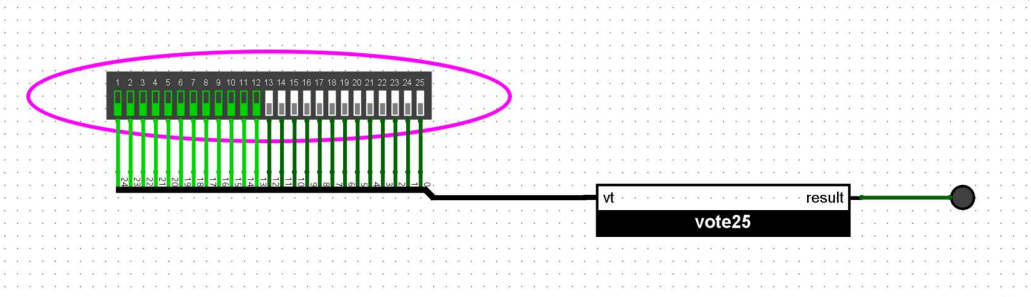
```
library ieee;
use ieee.std_logic_1164.all;

entity vote25 is
    port (
        vt     : in  std_logic_vector(25 downto 1);
        result : out std_logic
    );
end entity vote25;

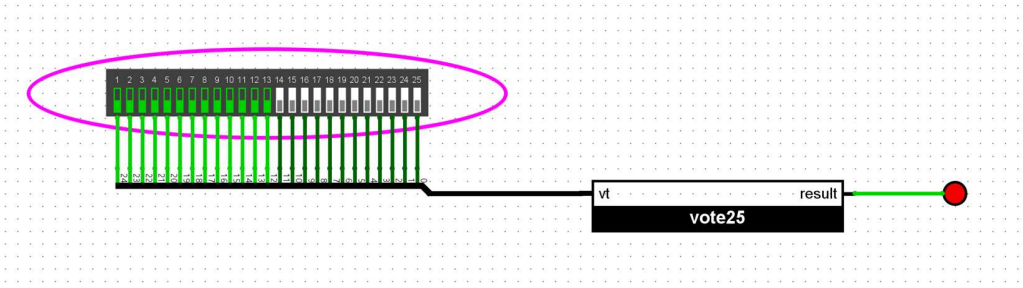
architecture bhv of vote25 is
begin
    process(vt)
        variable sum : integer range 0 to 25;
    begin
        sum := 0;
        for i in 1 to 25 loop
            if vt(i) = '1' then
                sum := sum + 1;
            end if;
        end loop;
        if sum >= 13 then
            result <= '1';
        else
            result <= '0';
        end if;
    end process;
end architecture bhv;
```

Problem 4 - Test results

Number of 1s	result
12	0
13	1



12 switches on -> LED off



13 switches on -> LED on

Problem 5. Count of 1s on 7-segment display

Counts the 1s in a 15-bit vt and shows 0-9 as digits and 10-15 as A-F (A, b, C, d, E, F). en=0 turns the display off.

Source: count_led.vhd

```
library ieee;
use ieee.std_logic_1164.all;

entity count_led is
    port (
        en      : in  std_logic;
        vt      : in  std_logic_vector(15 downto 1);
        display : out std_logic_vector(7 downto 0)
    );
end entity count_led;

architecture bhv of count_led is
begin
    process(en, vt)
        variable sum : integer range 0 to 15;
    begin
        sum := 0;
        for i in 1 to 15 loop
            if vt(i) = '1' then
                sum := sum + 1;
            end if;
        end loop;

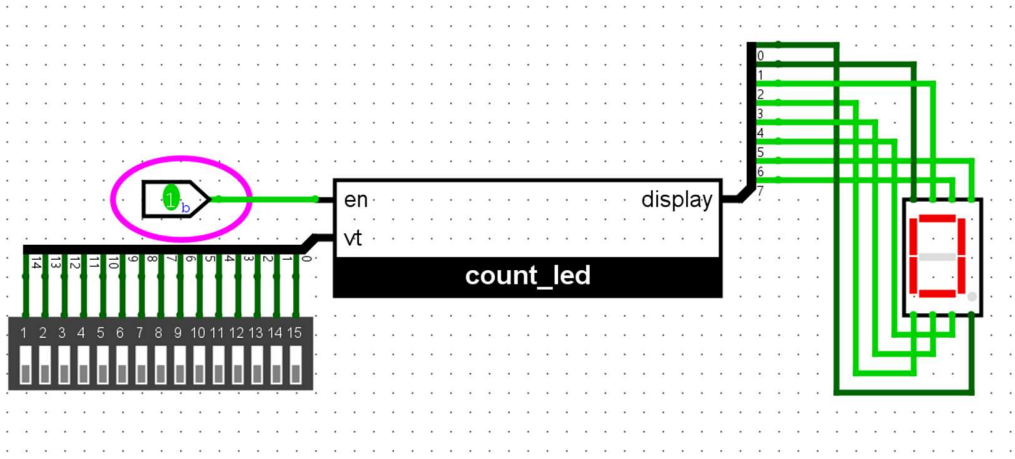
        if en = '0' then
            display <= "00000000";
        else
            case sum is
                when 0  => display <= "11111100";
                when 1  => display <= "01100000";
                when 2  => display <= "11011010";
                when 3  => display <= "11110010";
                when 4  => display <= "01100110";
                when 5  => display <= "10110110";
                when 6  => display <= "10111110";
                when 7  => display <= "11100000";
                when 8  => display <= "11111110";
                when 9  => display <= "11100110";
                when 10 => display <= "11101110"; -- A
                when 11 => display <= "00111110"; -- b
                when 12 => display <= "10011100"; -- C
                when 13 => display <= "01111010"; -- d
                when 14 => display <= "10011110"; -- E
                when 15 => display <= "10001110"; -- F
                when others => display <= "00000000";
            end case;
        end if;
    end process;
end architecture bhv;
```

Problem 5 - Test results

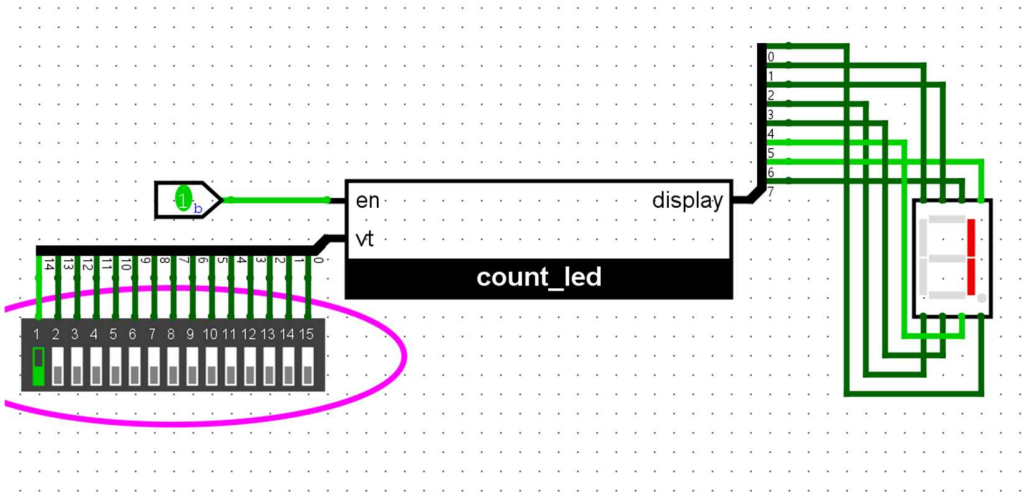
en	Number of 1s	Display
1	0..9	0..9
1	10	A
1	11	b
1	12	C
1	13	d

1	14	E
1	15	F
0	any	off

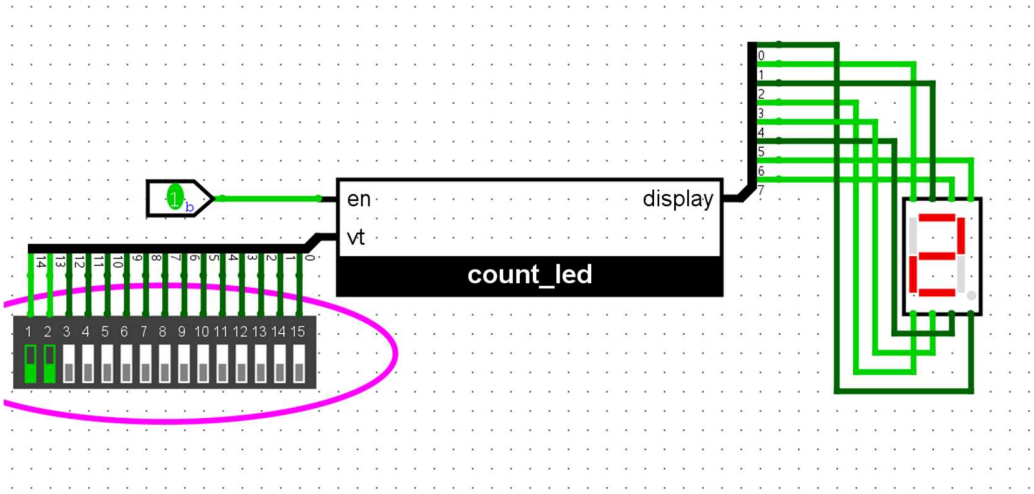
The task statement mentions a 13-bit vt, but a 15-bit vt is used so that counts 14 and 15 (E and F) can also be demonstrated.



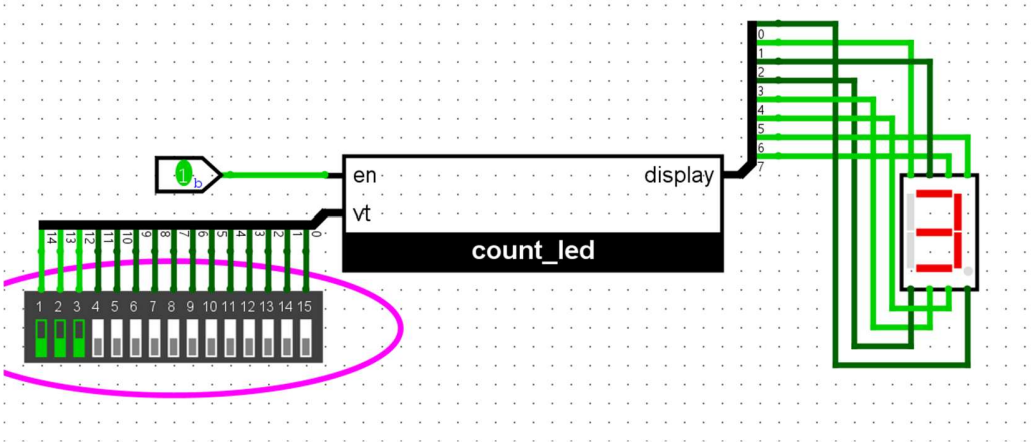
en=1, 0 switch(es) on -> 0



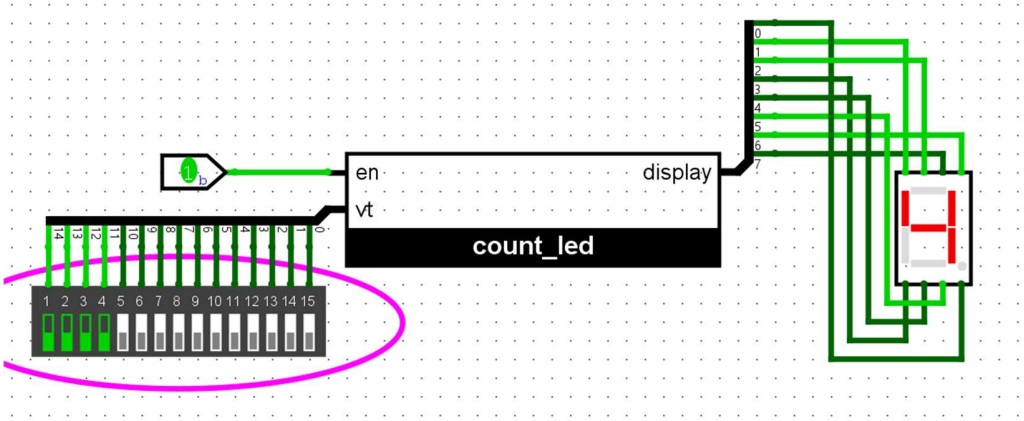
en=1, 1 switch(es) on -> 1



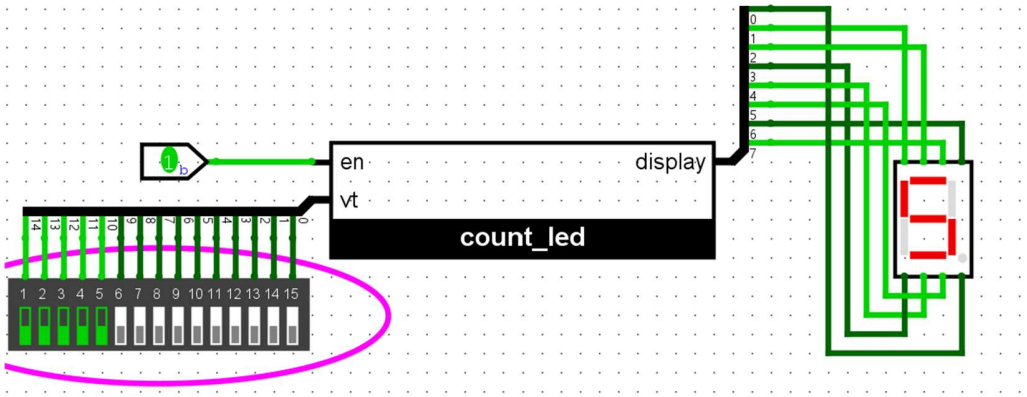
en=1, 2 switch(es) on -> 2



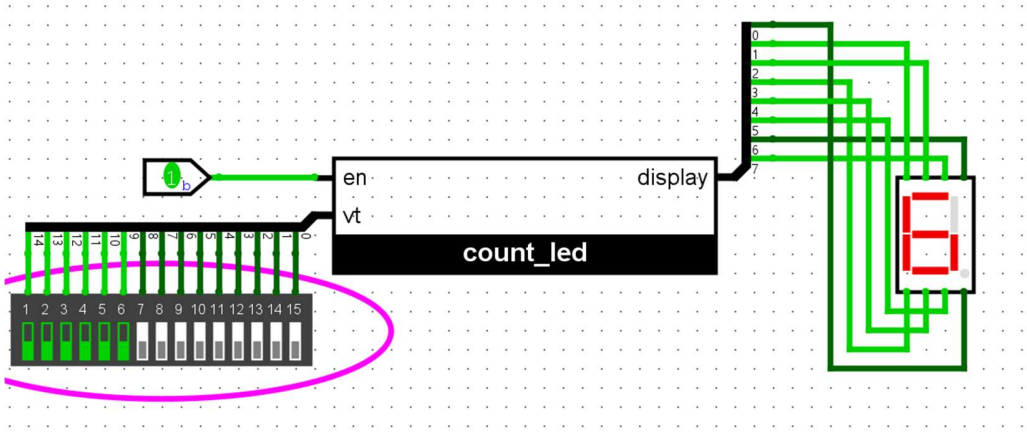
en=1, 3 switch(es) on -> 3



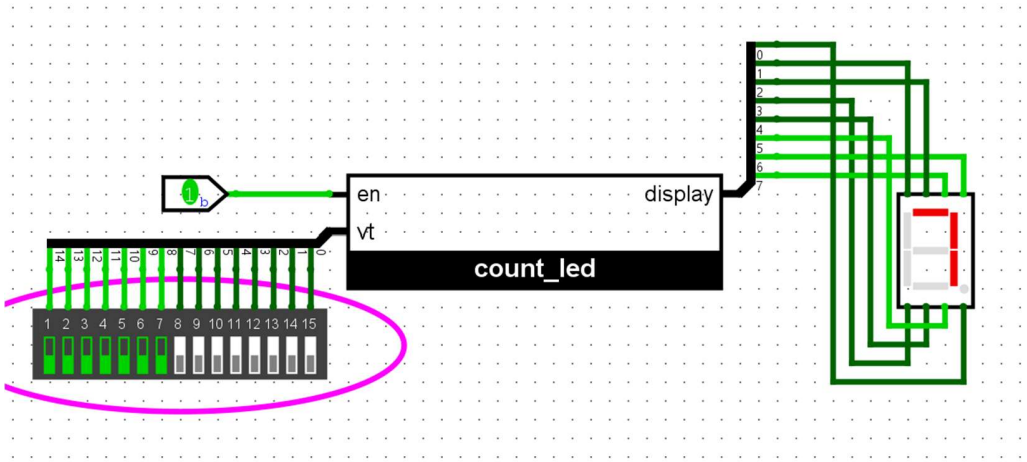
en=1, 4 switch(es) on -> 4



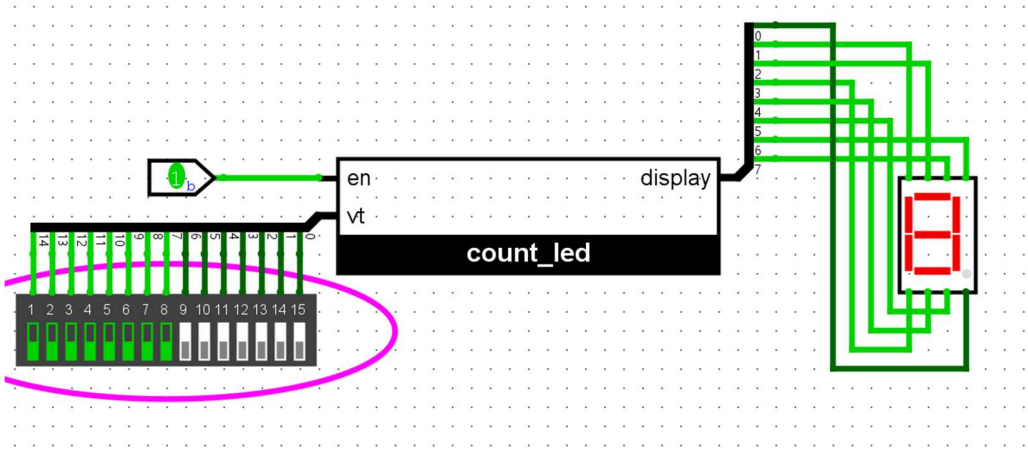
en=1, 5 switch(es) on -> 5



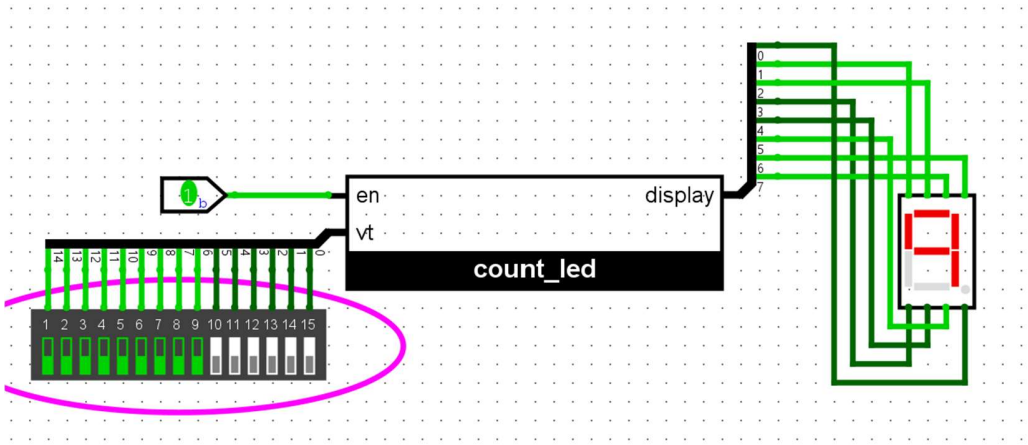
en=1, 6 switch(es) on -> 6



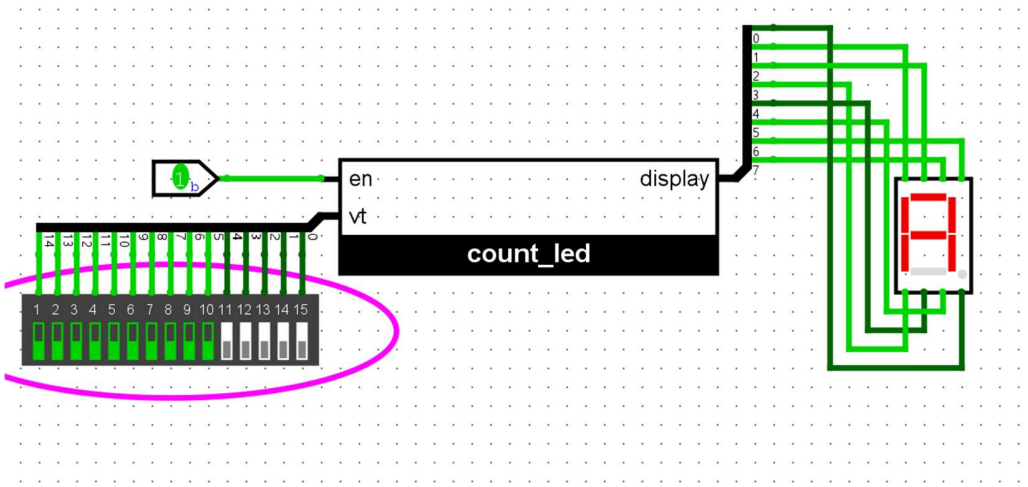
en=1, 7 switch(es) on -> 7



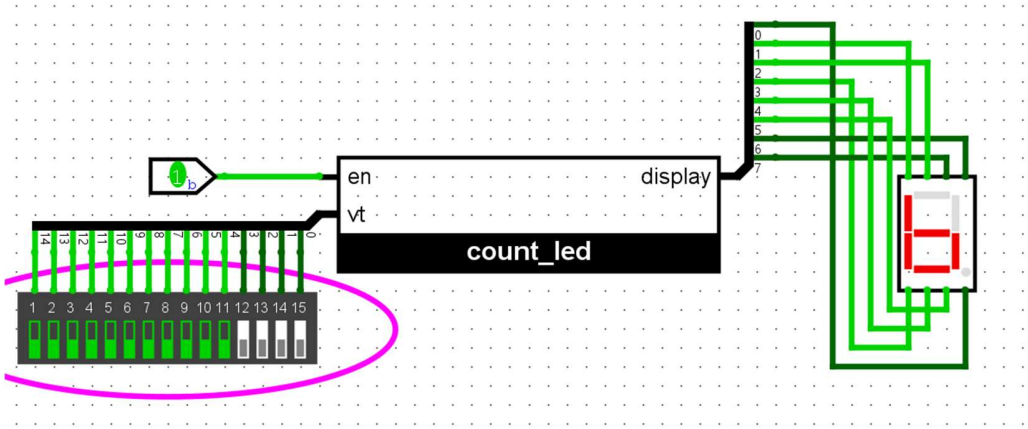
en=1, 8 switch(es) on -> 8



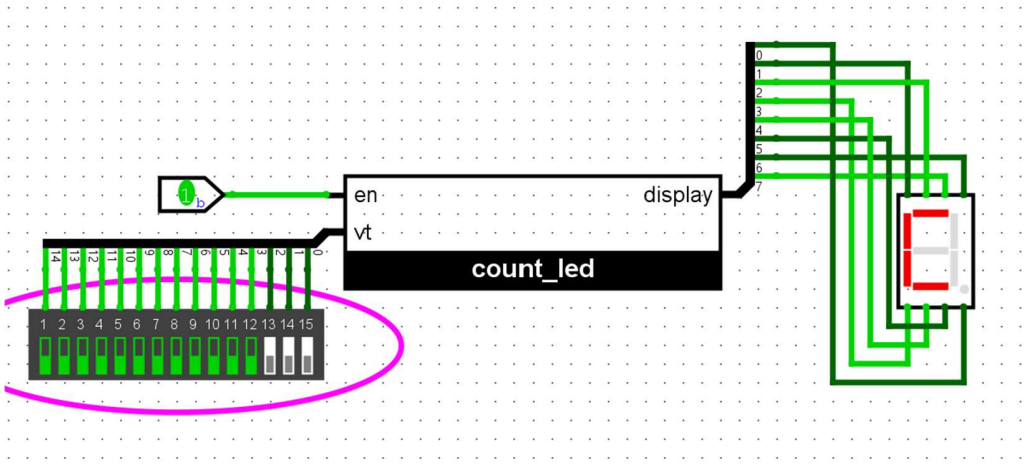
en=1, 9 switch(es) on -> 9



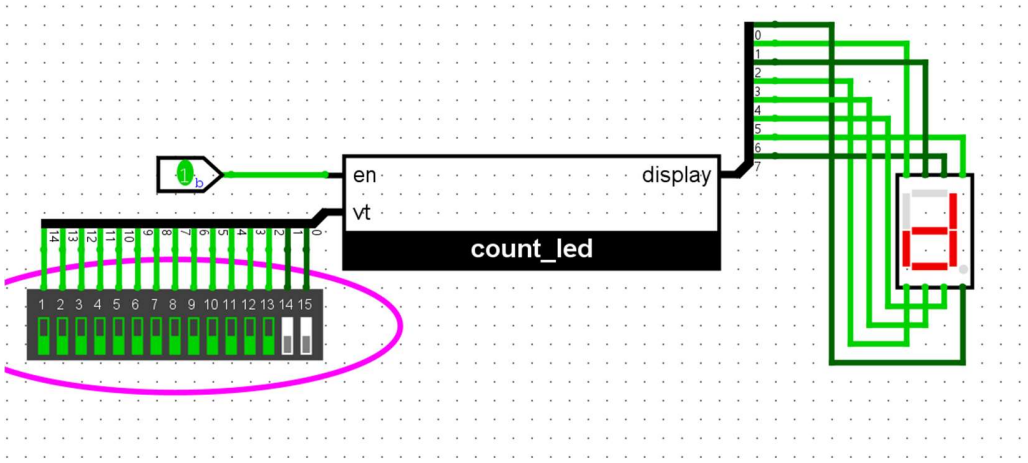
en=1, 10 switch(es) on -> A



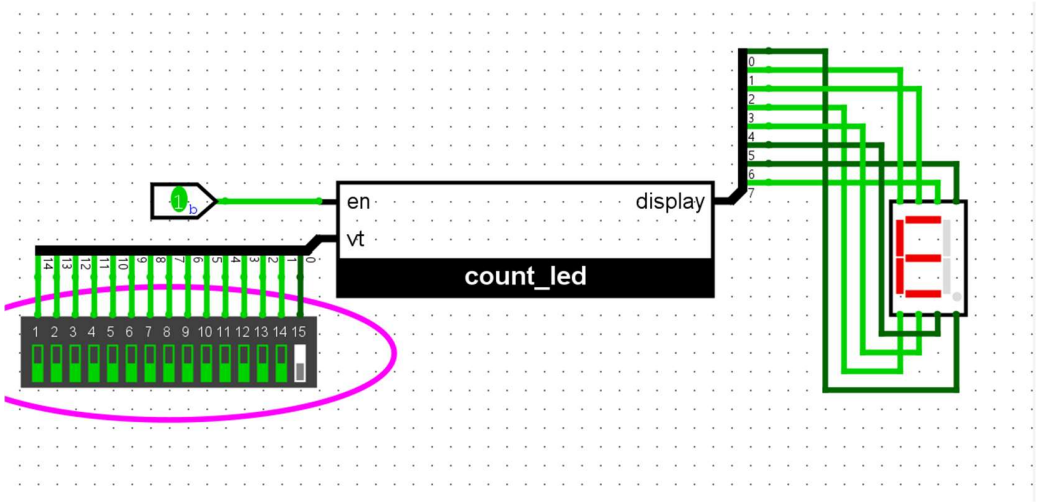
en=1, 11 switch(es) on -> b



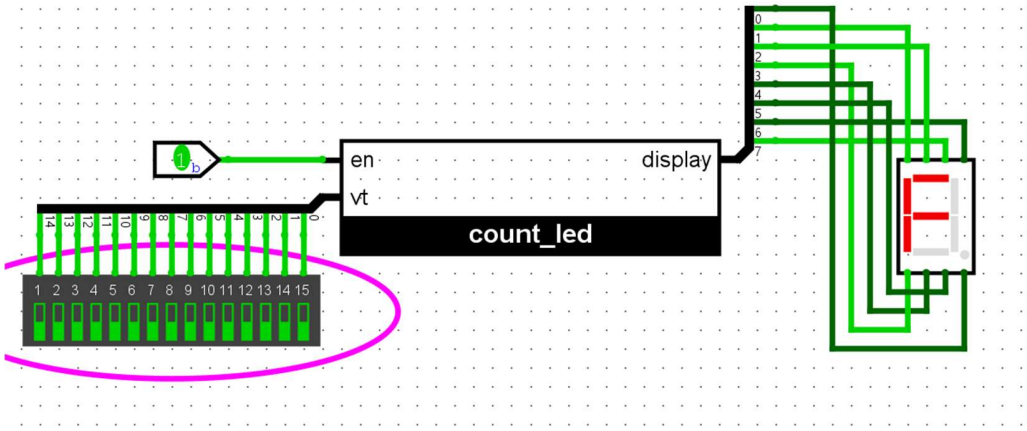
en=1, 12 switch(es) on -> C



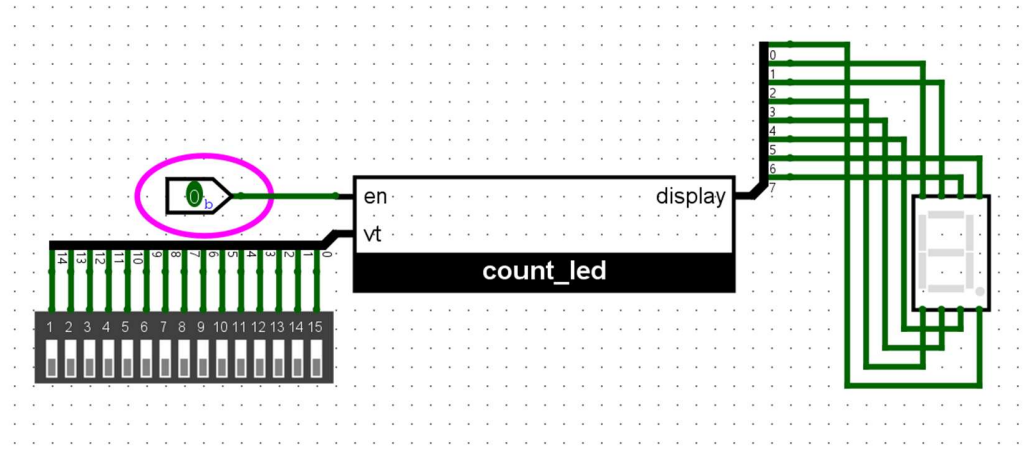
en=1, 13 switch(es) on -> d



en=1, 14 switch(es) on -> E



en=1, 15 switch(es) on -> F



en=0 -> display off